

it would be possible to deduce the equations which connect the new coefficients; but these are more easily obtained from the consideration that we must have identically

$$P_1Q_1 - Z_1^2 = PQ - Z^2;$$

the equations are thus found to be

$$\begin{aligned} A''^2 - AA' = 0, \quad B''^2 - BB' = 0, \quad C''^2 - CC' = 1, \\ 2A''B'' - AB' - A'B = -1, \quad 2A''C'' - AC' - A'C = 0, \quad 2B''C'' - BC' - B'C = 0. \end{aligned}$$

17. The general theory of the transformation of a quadric function into itself enables us to express the coefficients in terms of three arbitrary parameters. There is no difficulty in working out the formulae, and we finally obtain

$$\begin{aligned} \Omega P_1 &= -(\nu + 1)^2 P - \lambda^2 Q + 2\lambda(\nu + 1)Z, \\ \Omega Q_1 &= -\mu^2 P - (\nu - 1)^2 Q + 2\mu(\nu - 1)Z, \\ \Omega Z_1 &= -\mu(\nu + 1)P - \lambda(\nu - 1)Q + (-1 + \nu^2 + \lambda\mu)Z; \end{aligned}$$

and conversely

$$\begin{aligned} \Omega P &= -(\nu - 1)^2 P_1 - \lambda^2 Q_1 + 2\lambda(\nu - 1)Z_1, \\ \Omega Q &= -\mu^2 P_1 - (\nu + 1)^2 Q_1 + 2\mu(\nu + 1)Z_1, \\ \Omega Z &= -\mu(\nu - 1)P_1 - \lambda(\nu + 1)Q_1 + (1 + \nu^2 + \lambda\mu)Z_1, \end{aligned}$$

where $\Omega = -1 + \nu^2 - \lambda\mu$: it can be at once verified that each of the two sets of formulae does, in fact, give $P_1Q_1 - Z_1^2 = PQ - Z^2$.

18. The pseudosphere is a surface of revolution having for its meridian section the curve $x = \log \cot \frac{1}{2}\theta - \cos \theta$, $y = \sin \theta$. This is a curve symmetrical in regard to the axis of y ; and we obtain the portion of it lying on the positive side of this axis, by giving to θ the series of values $\theta = 0$ to $\theta = 90$; for $\theta = 0$, we have $y = 0$, $x = \infty$, or the axis of x is an asymptote; for $\theta = 90$, $x = 0$, $y = 1$, the point being a cusp of the curve. The geometrical definition is that the portion of the tangent included between the curve and the axis of x has the constant length = 1; the inclination of

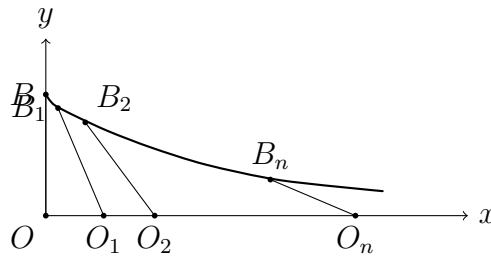


Fig. 2.

the tangent is in fact $= \theta$. We have $dx = \frac{\cos^2 \theta d\theta}{\sin \theta}$, $dy = \cos \theta d\theta$; and thence $ds = \cot \theta d\theta$, and the length in question is $\frac{y ds}{dy} = 1$. The curve may be constructed graphically: take (fig. 2) the distance $BO = 1$, on OB , B_1 very near to B , and then $B_1O_1 = 1$; on

O_1B_1 ; B_2 very near to B_1 , and then $B_2O_2 = 1$, and so on; the curve is shown on a larger scale in fig. 3, p. 235.

But the curve may also be laid down numerically; writing $\alpha = \frac{1}{2}\pi - \theta$, so that α is the inclination of the tangent to the axis of y , we have $x = \log \tan(\frac{1}{4}\pi + \frac{1}{2}\alpha) \sin \alpha$, $y = \cos \alpha$, where $\log \tan(\frac{1}{4}\pi + \frac{1}{2}\alpha)$, the hyperbolic logarithm (which has been the signification of $\log$ throughout), is the function tabulated Tab. IV., Legendre's *Traité des Fonctions Elliptiques*, t. II. pp. 256–259.

We may hence obtain the values of the coordinates as follows:

$\alpha = 90 - \theta$	0	10	20	30	40	50	60	70
x	∞	2.3400	1.2096	0.7154	0.4113	0.2110	0.0875	0.0242
y	1	0.9848	0.9397	0.8660	0.7660	0.6428	0.5000	0.3420
α	72	74	76	78	80	82	84	86
x	0.0160	0.0092	0.0045	0.0018	0.0005	0.0001	0.0000	0.0000
y	0.3090	0.2756	0.2419	0.2079	0.1736	0.1392	0.1045	0.0698

If $E = 0$, we have $A + C\dot{\phi} = 0$, that is, $\dot{\phi} = \text{const.}$, which belongs to the meridians; if B be not $= 0$, we may by a mere change of ϕ , that is, of the initial meridian, reduce the form to $A + B(\phi^2 + \text{cosec}^2 \theta) = 0$, which is the equation of a geodesic cutting at right angles the meridian $\phi = 0$; writing herein $\sin \theta = \frac{1}{r}$, we have $A + B(\phi^2 + \frac{1}{r^2}) = 0$, which is the equation in the polar coordinates r, ϕ of the projection of the geodesic on the equatorial plane x ; putting herein for greater convenience $B = Ak^2$, we have $\frac{1}{r^2} = \frac{k^2}{1 - k^2\phi^2}$: we require only such portions of the curves as lie within the unit-circle, and need therefore attend only to those for which k is not greater than 1.

19. And in any such curve consider ϕ as extending from $\phi = 0$ to $\phi = \gamma$, writing K this last value $= \gamma$, we have $k = \frac{\gamma}{\sqrt{1 + \gamma^2}}$; if $\gamma < \pi$, that is, $k < \frac{\gamma}{\sqrt{1 + \pi^2}}$, the curve is a mere arc cutting at right angles (at the distance $r = k$ from the centre) the meridian $\phi = 0$, and extending itself out on each side to meet the unit-circle in the points $\phi = \gamma, \phi = -\gamma$ respectively; in the case $\gamma = \pi$, that is, $k = \frac{\pi}{\sqrt{1 + \pi^2}}$, the two points $\phi = \gamma$ come together

at the point $\phi = \pi$, or the curve becomes a loop; and for larger values, $k = \frac{2\pi}{\sqrt{1 + 4\pi^2}}$ to $k = \frac{3\pi}{\sqrt{1 + 9\pi^2}}$, we have the two branches crossing: each other on the meridian $\theta = \pi$

at the distance $r = \frac{k}{\sqrt{1 - k^2\pi^2}}$ from the centre and then extending themselves in the opposite semicircles, so as to meet the unit-circle at the points $\phi = \gamma$. And we have thus another critical value $k = \frac{2\pi}{\sqrt{1 + 4\pi^2}}$, for which the two branches having thus crossed each other come to unite themselves at the point $\phi (= 2\pi) = 0$ of the unit-circle; and in like manner the critical values $\frac{3\pi}{\sqrt{1 + 9\pi^2}}, \frac{4\pi}{\sqrt{1 + 16\pi^2}}, \&c.$: for a value of k between such limits, the branch is a spiral having a determinate number of convolutions, and the two branches cross each other always on the radii $\phi = 0$ and $\phi = \pi$ respectively.

20. Let ψ denote the inclination of the radius vector to the normal, or, what is the same thing, that of the element of the circular arc to the tangent; we have $\tan \psi = \frac{dr}{r d\phi}$, and

$\frac{dr}{r d\phi} = \frac{k^2\phi}{r^2}, = r^2\phi$, that is, $\tan \psi = r^2\phi$. At the intersection with the unit section $r = 1$, and therefore $\tan \psi = \phi$; moreover putting $\theta = \cos K$, so that the equation of the curve now is $r^2 = \frac{\theta}{\theta - \cos^2 K}$, then for $r = 1$ we have $\theta = \tan K; 1 - \phi^2 \cos^2 K$

and therefore $\tan K = \pi$, we have $K = 72\ 20'$ nearly: the complement hereof $17\ 40'$ is thus the angle at which each branch of the loop cuts the meridian $\phi = \pi$.

21. To obtain another datum convenient in tracing the curve, I write $\phi_0 = \tan K$ for the value of ϕ at the unit-circle; and introducing for the moment the rectangular coordinates $X = r \sin \phi$, $Y = 1 - r \cos \phi$, then we easily find

$$\frac{dY}{dX} = \frac{r \sin \phi_0 - \cos \phi}{r \cos \phi_0 + r^3 \sin \phi_0 \sin \phi},$$

and thence, for the equation of the tangent at the point on the unit-circle,

$$\sin \phi_0 \cos \phi \cdot (Y - 1 + \cos \phi_0) = -\frac{1}{r^2}(X - \sin \phi_0 \cos \phi_0).$$

For the tangent at the point of intersection with the radius $\phi = 0$, or say the *apse*, we have $y = 1 - \cos K$; and hence, at the intersection of the two tangents,

$$x = \sin \phi_0 + \frac{\cos \phi_0 + \sin \phi_0}{\sin \phi_0 \cos \phi_0}(\cos \phi - \cos K),$$

which, putting therein $\phi_0 = \tan K$, becomes

$$\cos K \{1 - \cos(\phi - K)\} = \frac{x}{\sin(\phi - K)} = \cos K \tan \frac{1}{2}(\phi - K),$$

where ϕ is given in terms of ϕ_0 by the just-mentioned equation $\phi_0 = \tan K$.

We have $y = 1 - \cos K$, $x = \cos K \tan \frac{1}{2}(\phi - K)$, for the locus of the intersection of the two tangents; this is easily seen to be a curve having a cusp at the unit-circle.

22. Fig. 3 shows the curves for the values $\phi_0 = \tan K$

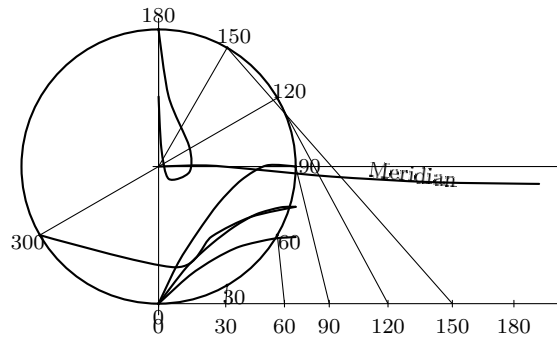


Fig. 3.

$\phi_0 = 90$, we join with the centre of the circle the point 90 of the tangent, thus determining on the unit-circle a point belonging to the angle $K = 57\,31'$; at this point we draw parallel to the tangent a line which is the tangent at the lowest point; the curve passes through the point 90 on the unit-circle, and there cuts the circle at the angle $\kappa = 57\,31'$ (or, what is the same thing, the radius at the complementary angle), and we have thus the tangent at the point 90 of the unit-circle; it will be noticed that this meets the tangent at the apse at a point near to this apse, so that the arc as determined by the two tangents is for a large part of its course nearly a right line; this is still more the case for smaller values of ϕ_0 or κ , while for larger values the deviation increases, but in the neighbourhood of the unit-circle the form is always nearly rectilinear.

I show in the same figure the form of the curve for $\phi_0 = 300, = 5.2359877, = \tan \kappa$, that is, $\kappa = 79\,11'$, $r = \cos \kappa = 0.1876670$, the value at the apse: the construction for the tangent at the unit-circle is the same as before, but in order to lay down the curve with tolerable accuracy we require also the value of r at the node on the meridian $\phi = 180$; this is, of course, given by $r = \frac{\cos \kappa}{\sqrt{1 - \pi^2 \cos^2 \kappa}}$, that is, $r = \frac{\cos \kappa}{\cos \alpha}$, if $\pi \cos \kappa = \sin \alpha$; whence without difficulty $r = 0.23236$, the value at the node.

23. The curves shown in the figure are projections upon the plane of the unit-circle, viz. they are the projections on this plane of the geodesics, which cut at right angles a given meridian; but bearing in mind the form of the meridian, it is easy, by means of the projection, to understand the actual forms on the surface of the pseudosphere. A point near the centre of the figure represents a point high up on the surface; and in any radius the portions near the centre are the more foreshortened in the figure, and represent greater distances on the surface. Each geodesic cutting the meridian at right angles at the apse descends symmetrically on the two sides, reaches ultimately—it may be after many convolutions—the unit-circle; the meridian itself is a limiting or special form of geodesic. The unit-circle is not properly a geodesic, but it is an envelope of geodesics.

24. To obtain all the geodesics, we have to consider the geodesics which cut at right angles a given meridian; and then to imagine this meridian (with the geodesics which belong to it) turned round so as to occupy successively the positions of all the other meridians. The same remark applies of course to the projections; the figure shows the projections cutting at right angles a given radius of the circle; and this radius (with the projections belonging to it) is then to be turned round so as to occupy successively the positions of all the other radii. We may imagine the several geodesics turned round separately, each through a different angle, so as to bring each of them to pass through one and the same point of the surface; we have then the geodesics drawn in all directions through this point of the surface; doing the same thing with the projections, we have, it is clear, the projections of the geodesics drawn in all directions through the point. It is easy, by drawing the projections each on a separate circle of paper, and passing a pin through the centres, to form a model by means of which an accurate figure of the projections of the geodesics through any given point can at once be exhibited.

25. Taking Q a given point, and EG the radius (or meridian) through Q , then the projections through Q are symmetrically situate in regard to this radius EG ; the curve which has its apse at Q is the projection which cuts the radius at right angles; the projections which lie outside this curve, that is, on the side towards E , meet the unit-circle on the two sides of E , in points A, D , belonging each of them to a positive value of κ , which may be anything from 0 to 90; the projections which lie inside the curve meet

the unit-circle in points B, C , belonging to the values of κ for which ϕ_0 lies between 180 and 360, and are therefore curves which cross the radius EG .

There are in the whole four projections through Q which touch the unit-circle, corresponding to the values $\phi_0 = 180$, $\phi_0 = 360$, and the coincident values $\phi_0 = 0 = 720$; in the case $\phi_0 = 180$ (or what is the same thing, $\kappa = 90$, the apse is at the distance $r = 0$ from the centre), the curve is a loop touching the unit-circle at the extremity E of the radius EG , and the branch on the same side with the apse is really a distinct curve intersecting the loop; in the case $\phi_0 = 360$ the curve is a loop touching at the other extremity G ; and in the case $\phi_0 = 0 = 720$, the curve is the unit-circle itself; or, more accurately, the geodesics become the unit-circle at opposite sides of the surface, meeting each other at the cusp; in the model there is a break, or rather we pass continuously from one circle to the other at opposite points of the unit-circle, or say at the points G and E on the different circles of the model.

The outside projections through Q divide the portion outside the curve, apse at Q , into separate regions; but these projections (taken the whole way from Q to the unit-circle) do not each of them form a single geodesic; any such projection is really the projection of two geodesics, viz. a geodesic from Q to the unit-circle at A , and a different geodesic from Q to the unit-circle at D ; the model shows the course of these geodesics from Q to the unit-circle, each as a branch of a geodesic which has its apse on the radius EG produced beyond Q . It is to be added that these outside projections, each as belonging to two geodesics, do not each of them constitute a boundary between two regions; the case is really the same as that of a meridian on the sphere in regard to the geodesics through a point Q of it: the meridian itself is a geodesic through Q , and it is also the limiting form of geodesics on each side of it; and in the projection it divides the regions on the two sides of it, but it does not constitute a boundary between two distinct regions.

The inside projections through Q are (on account of the crossing) much more complicated; it would require a careful examination of the model to understand their true form and connexions.

26. It is interesting to connect the theory of the geodesics of the pseudosphere with the general theory of geodesics. Starting with the form

$$ds^2 = \cot^2 \theta d\theta^2 + \sin^2 \theta d\phi^2,$$

$= E d\theta^2 + 2F d\theta d\phi + G d\phi^2$, we have $E = \cot^2 \theta$, $F = 0$, $G = \sin^2 \theta$; and therefore $E + 1 = \frac{1}{G}$, or $E = \frac{1}{G} - 1$, and the differential equation of the geodesic becomes

$$\phi' \{2GE' - GE_1\} \theta' + GG_2 \phi'^2 + 2EG(\theta' \phi'' - \theta'' \phi') = 0,$$

that is,

$$\phi' \{(2EG_1 - GE_1)\theta'^2 + GG_2 \phi'^2\} + 2EG(\theta' \phi'' - \theta'' \phi') = 0,$$

where $E_1 = \frac{dE}{d\theta}$, $G_1 = \frac{dG}{d\theta}$, $G_2 = \frac{dG}{d\phi}$; and writing herein

$$E = \frac{1-G}{G}, \quad E_1 = -\frac{G_1}{G^2},$$

we have

$$2EG_1 - GE_1 = \frac{2(1-G)}{G} G_1 + \frac{G_1}{G} = \frac{(3-2G)G_1}{G}.$$

Moreover, from $G = \sin^2 \theta$, we find $G_1 = 2\sqrt{G(1-G)}$; and the equation becomes

$$\phi' \left\{ \frac{(3-2G)G_1}{G} \theta'^2 + GG_2 \phi'^2 \right\} + 2(1-G)(\theta' \phi'' - \theta'' \phi') = 0.$$

Introducing here G in place of θ by the equation $G = \sin^2 \theta$, we have $\frac{dG}{d\theta} = G_1$, $\frac{d^2 G}{d\theta^2} = G_{11}$, and the equation thus becomes

$$\phi' \left\{ (3-2G) \frac{dG}{d\theta} \theta'^2 + G^2 G_2 \phi'^2 \right\} + 2G(1-G)(\theta' \phi'' - \theta'' \phi') = 0.$$

the arc EG . This of course corresponds to the Lobatschewskian theory, according to which we have through a point Q to the extremities at infinity of a line EG , two distinct lines QB, QC , said to be the *parallels* through Q of the line BC ; and which are such that any line through Q intermediate to QB, QC meets the line BC ; while any line through Q extramediate to QB, QC does not meet the line BC .

27. Reverting to the equations of No. 4, these may be written

$$\begin{aligned}\Omega P_1 &= -(\nu + 1)^2 P - \lambda^2 Q + 2\lambda(\nu + 1)Z, \\ \Omega Q_1 &= -\mu^2 P - (\nu - 1)^2 Q + 2\mu(\nu - 1)Z, \\ \Omega Z_1 &= -\mu(\nu + 1)P - \lambda(\nu - 1)Q + (-1 + \nu^2 + \lambda\mu)Z.\end{aligned}$$

Taking P, Q, Z as the coordinates of a point in ordinary space, and P_1, Q_1, Z_1 as those of another point, these equations establish a correspondence between the two points; to any given position of either point there corresponds a single position of the other point. Starting from the quadric surface $PQ - Z^2 = 0$, this is transformed into the quadric surface $P_1Q_1 - Z_1^2 = 0$; and conversely. In particular, the tangent plane at any point of either quadric corresponds to the tangent plane at the corresponding point of the other quadric. But the whole theory is really included in the less general theory of the linear transformation of a quadric function into itself; and I do not at present enter into it any further.

Cambridge, England, 17th March, 1884.

A MEMOIR ON SEMINVARIANTS.

[From the *American Journal of Mathematics*, t. VII. (1885), pp. 1—25.]

Introductory. Art. Nos. 1 to 3.

1. A VERY remarkable discovery in the Theory of Seminvariants has been recently made by Capt. MacMahon, viz. considering the equation

$$0 = 1 + b\frac{x}{1} + c\frac{x^2}{1.2} + d\frac{x^3}{1.2.3} + \&c.,$$

and its roots $\alpha, \beta, \gamma, \dots$, as defined by the identity

$$1 + b\frac{x}{1} + c\frac{x^2}{1.2} + \&c. = 1 - \alpha x \cdot 1 - \beta x \cdot 1 - \gamma x \dots,$$

then, any symmetrical function of the roots being represented by a partition symbol in the usual manner, $1 = \Sigma\alpha$, $2 = \Sigma\alpha^2$, 11 or $1^2 = \Sigma\alpha\beta$, &c., the theorem is that any symmetric function represented by a non-unitary symbol (or symbol not containing a 1), say for shortness any non-unitary symmetric function, is a seminvariant in regard to the coefficients $1, b, c, d, e$, &c.

We have for instance

$$\begin{aligned} 2 &= -(c - b^2), \\ 3 &= -\frac{1}{2}(d - 3bc + 2b^3), \\ 4 &= \frac{1}{6}(-e + 4bd + 3c^2 - 12b^2c + 6b^4), \end{aligned}$$

and so on.

2. The proof is very simple; writing $x = y - b$, the equation becomes

$$0 = 1 + b\frac{(y-b)}{1} + c\frac{(y-b)^2}{1.2} + d\frac{(y-b)^3}{1.2.3} + \&c.,$$

or, what is the same thing,

$$0 = A + By + Cy^2 + Dy^3 + \&c.,$$

where A, B, C, D , &c. are functions of b, c, d, e , &c.; then the coefficients A, B, C, D , &c. are seminvariants, and therefore also any function of these coefficients which does not contain the absolute coefficient A ; that is, any non-unitary symmetric function of the roots of the equation in y , or, what is the same thing, of the roots of the original equation in x .

where to verify that this is a seminvariant, observe that the equation in y is $0 = A + By + Cy^2 + Dy^3 + \&c.$, and that the coefficients $C, D, E, \&c.$ (the coefficients after B) are seminvariants; and further that A does not enter into the expression of the non-unitary symmetric function.

3. To further explain the notion of a non-unitary symmetric function, observe that by the theorem the number of seminvariants of a given weight is equal to the number of non-unitary partitions of that weight; thus for weight = 6 the partitions are

$$6, \quad 42, \quad 33, \quad 222, \quad 52, \quad 43, \quad 322, \quad 222,$$

eight in all, which is the number of the seminvariants of the weight 6; but among these there is only one partition into two equal parts, viz. 33, and there is thus only one seminvariant of the weight 6 which is a perfect square. Seminvariants which are not non-unitary symmetric functions are called by Sylvester *Subinvariants*; they are symmetric functions which contain a unit: the numbers are for weight = 6 the partitions 51, 411, 321, 3111, 2211, 21111, 111111, seven in all, and these are the subinvariants of the weight 6. The whole number of partitions of 6 is 11, which is the number of seminvariants + subinvariants, that is, of symmetric functions of the weight 6. In the case of a subinvariant the expression as a function of the coefficients will contain A (and will in fact be a rational and integral function of $A, B, C, D, E, F, G, \&c.$).

Definition of Seminvariant. Art. Nos. 4 to 6.

4. I give the definition of a seminvariant. Starting from the binary quantic

$$(a, b, c, d, \dots \backslash x, y)^n,$$

writing for convenience $(a, b, c, d, \dots)$ to denote the coefficients (taking them to be independent of each other), then a seminvariant is a function of the coefficients which is reduced to zero by the operation

$$a\partial_b + 2b\partial_c + 3c\partial_d + \dots,$$

or, what is the same thing, it is a function of the coefficients $(a, b, c, d, \dots)$ which is reduced to zero by the operation

$$\delta = a\partial_b + 2b\partial_c + 3c\partial_d + \dots + nb\partial_l,$$

or, what is again the same thing, it is a function S of the coefficients, such that we have

$$(a, b, c, d, \dots \backslash \partial_y, -\partial_x)(a, b, c, d, \dots \backslash x, y)^n \cdot S = 0.$$

5. We may, taking in the first instance $(a, b, c, \dots)$ as the coefficients of the several powers of x , write

$$0 = 1 + b\frac{x}{1} + c\frac{x^2}{1.2} + d\frac{x^3}{1.2.3} + \&c.,$$

where 1 is put for a , or, what is the same thing, we may write

$$(a, b, c, d, e, f, \& 1, x)^6 = 0,$$

or for the general case

$$(a, b, c, d, \dots \setminus 1, x)^n = 0.$$

The seminvariants of this equation are the same as those of the binary quantic $(a, b, c, d, \dots \setminus x, y)^n$ of the order n ; only we consider only the seminvariants of a degree in the coefficients which does not exceed n , and only those of a weight which does not exceed the weight of the last coefficient l , that is $\frac{1}{2}n(n-1)$.

6. Observe that, if the equation in x be $0 = (a, b, c, \dots \setminus 1, x)^n$, then the equation in y obtained by the change $x = y - \frac{b}{a}$ is $0 = (a, B, C, D, \dots \setminus 1, y)^n$, where the new coefficients B, C, D , &c. are given functions of a, b, c , &c.; the seminvariants are the same for the two equations, but for the equation in y the coefficient B vanishes; any seminvariant of the equation in x is thus a seminvariant of the equation in y , and conversely any seminvariant of the equation in y is a seminvariant of the equation in x ; the seminvariants of the equation in y are functions of a, C, D, E , &c. which are reduced to zero by the operation $2C\partial_D + 3D\partial_E + \dots$; or, what is the same thing, the seminvariants of the equation in x are functions of the original coefficients a, b, c , &c. which are reduced to zero by the operation

$$\delta = a\partial_b + 2b\partial_c + 3c\partial_d + \dots,$$

and which, when for b is substituted $-\frac{B}{a}$, become functions of a, C, D, E , &c. which are reduced to zero by the operation $2C\frac{\partial}{\partial D} + 3D\partial_E + \dots$.

The Seminvariants are the Non-unitary Symmetric Functions.

Art. Nos. 7 to 12.

7. In forming a product $l.m.n.\dots p.q.r.\dots$, we may have equalities among the numbers $l, m, n, \dots$, or among the numbers $p, q, r, \dots$, or between numbers of the one set and numbers of the other set. Thus in $\{4, 3\}$, we have

$$\begin{aligned}\{4\} &= 4, & 31, & 22, & 211, & 1111, \\ \{3\} &= 3, & 21, & 111,\end{aligned}$$

and thence

$$\begin{aligned}\{4, 3\} &= 43, & 421, & 4111, & 331, & 3211, & 31111, \\ & 2221, & 22111, & 211111, & 1111111,\end{aligned}$$

ten terms in all, agreeing with the formula $\{4, 3\} = \{7\} + \{6, 1\} + \{5, 2\} + \{4, 3\}$, where the symbols on the right-hand denote the partitions of 7 into one, two, three, and four parts respectively; and where, to verify the number of terms, we have $1 + 3 + 3 + 3 = 10$.

The several terms are represented by their symbols, and the partition symbols are arranged in the order of the large numbers; when two symbols contain the same large numbers, the order is determined by the next largest numbers; and so on.

8. The process for finding the terms of $\{4, 3\}$ is as follows: the first term is obtained by adding the numbers of the second row successively to the corresponding numbers of the first row; viz. $4 + 3 = 7$ gives the first term 7; but $3 + 1 = 4$ would not give a second term, for the rule is that the partial sum must be greater than the corresponding number of the second row; $2 + 1 = 3$, and 3 being not greater than 3 (the corresponding number of the second row), we do not obtain a term; $1 + 1 = 2$, and we have the second term 42; and $1 + 0 = 1$, and we have the third term 411. We then add the numbers of the second row to the corresponding numbers of the first row, beginning always from the first number of the second row; we thus obtain $4 + 2 = 6$, and since the next number of the first row is 1, we add it to 1 obtaining $1 + 1 = 2$, which gives the term 33; we do not obtain any further term from this, since $1 + 1 = 2$ is not greater than 1 (the next number of the second row); and proceeding in like manner we obtain all the remaining terms.

9. For the proof of MacMahon's theorem, I consider a symmetric function represented by a partition symbol, and I show that if the symbol is non-unitary, then the function is a seminvariant; that is, that it is reduced to zero by the operation $\delta = a\partial_b + 2b\partial_c + 3c\partial_d + \dots$. The proof depends on the expression of the symmetric function as a sum of powers; using the series $s_1 = \Sigma\alpha$, $s_2 = \Sigma\alpha^2$, $s_3 = \Sigma\alpha^3$, &c., we have between these and the coefficients a , b , c , &c. the well-known relations

$$\begin{aligned} as_1 + b &= 0, \\ as_2 + bs_1 + 2c &= 0, \\ as_3 + bs_2 + cs_1 + 3d &= 0, \\ &\vdots \end{aligned}$$

whence, differentiating,

$$\begin{aligned} a\partial_b s_1 + 1 &= 0, \\ a\partial_b s_2 + s_1 &= 0, \\ a\partial_b s_3 + s_2 &= 0, \\ &\vdots \end{aligned}$$

and thence

$$\begin{aligned} \delta s_1 &= -1, \\ \delta s_2 &= -2s_1, \\ \delta s_3 &= -3s_2, \\ &\vdots \end{aligned}$$

or generally $\delta s_n = -ns_{n-1}$, where $s_0 = 1$.

10. If the partition symbol contain any number 1, say the numbers are 1, m , n , &c., then the symmetric function contains the factor s_1 , and we have $\delta(s_1 \cdot s_m \cdot s_n \dots) = s_m s_n \dots \delta s_1 + s_1 \delta(s_m s_n \dots) = -s_m s_n \dots + s_1 \delta(s_m s_n \dots)$, which is not $= 0$; the function is therefore in this case not a seminvariant. But if the symbol is non-unitary, that is, if it does not contain the number 1, then starting from the highest number p , the function is $s_p \cdot s_q \cdot s_r \dots$, and we have $\delta(s_p \cdot s_q \cdot s_r \dots) = s_q s_r \dots \delta s_p + s_p s_r \dots \delta s_q + \dots$; or, what is the same thing, the operation δ performed on the product gives a sum of terms each containing a factor $\delta s_n = -ns_{n-1}$; the number which accompanies this factor s_{n-1} is one of the original numbers, and therefore is not $= 1$; the number $n - 1$ is thus not $= 0$, and the symbol still contains only numbers different from 0. Continuing the operation δ , we always obtain terms in which the factors s have indices different from 0, and the process comes to an end only when one of the indices becomes $= 1$; but in this case the term contains the factor s_1 multiplied by a term having a numerical coefficient, and this coefficient is always $= 0$; the symmetric function is thus reduced to zero by the operation δ , that is, it is a seminvariant.

11. The proof is thus shown to depend on the law of the numerical coefficients; the law may be verified directly in particular cases. For the symmetric function $2 = s_2$, we have $\delta s_2 = -2s_1$, $\delta^2 s_2 = -2\delta s_1 = 2$, and thence $\delta^3 s_2 = 0$; whence 2 is a seminvariant. For $3 = s_3$, we have $\delta s_3 = -3s_2$, $\delta^2 s_3 = -3\delta s_2 = 6s_1$, $\delta^3 s_3 = 6$, and $\delta^4 s_3 = 0$; whence 3 is a seminvariant. For $22 = s_2^2$, we have $\delta(s_2^2) = 2s_2\delta s_2 = -4s_2s_1$, $\delta^2(s_2^2) = -4(s_1\delta s_2 + s_2\delta s_1) = 4(2s_1^2 + s_2)$, and it may be verified that the further operation δ reduces this to zero.

12. The proof may be presented in a different form; taking the symmetric function to be $p.q.r.\dots$, this is a function of the roots $\alpha, \beta, \gamma, \dots$; performing on it the operation $\delta = a\partial_b + 2b\partial_c + 3c\partial_d + \dots$, the result is the same as if we performed on it the operation $-\Sigma\partial_\alpha$; that is, we have $\delta(p.q.r.\dots) = -\Sigma\partial_\alpha(p.q.r.\dots)$. Now $\Sigma\partial_\alpha(\alpha^p \cdot \beta^q \cdot \gamma^r \dots)$, where the summation extends to all the roots, is equal to $p\alpha^{p-1}\beta^q\gamma^r \dots + q\alpha^p\beta^{q-1}\gamma^r \dots + \dots$; and the terms of this sum are all of them different from zero except when one of the indices $p, q, r, \dots$ becomes $= 0$; but in this case the term contains a factor which is a function of the roots in which one of the roots does not appear, and the summation Σ gives a term multiplied by the number of roots, which is infinite; the proper interpretation is that the coefficient in question vanishes; and we thus see that the operation δ reduces the symmetric function to zero.

Expression of Seminvariants in terms of the Coefficients.

Art. Nos. 13 to 16.

13. The seminvariant $2 = -(c - b^2)$; we may write this $2 = (b, \mathfrak{c} - b, 1)^2$; in fact, this is $(-b)^2c + (-b)(1)c + (1)^2c$, that is, $c - b^2 - b^2 + b^2 = c - b^2$, and the sign being changed, this is $-(c - b^2)$; or observing that $b^2 - c$ is the discriminant of $(b, \mathfrak{c} X, Y)^2$, the seminvariant 2 is the discriminant of $(b, \mathfrak{c} X, Y)^2$.

14. The seminvariant $3 = -\frac{1}{2}(d - 3bc + 2b^3)$; we may write this

$$3 = \frac{1}{2}(b, c, \mathfrak{d} - b, 1)^3,$$

or, what is the same thing,

$$3 = \frac{1}{1.2}(b, c, \mathfrak{d} - b, 1)^3.$$

15. The seminvariant 4 is the value of $(b, c, d, \mathfrak{d} - b, 1)^4$ divided by 1.2.3, as is seen by the actual development.

16. And generally a non-unitary symmetric function $p.q.r.\dots$ is equal to a numerical multiple of the seminvariant which is the discriminant of $(b, c, d, \dots, \mathfrak{d} - b, 1)^p$, or which is obtained by the development of this discriminant.

17. In explanation as to the discriminant of $(b, c, d, \dots - b, 1)^n$, observe that writing the quantic under the form

$$X^n b + nX^{n-1}Yc + \frac{1}{2}n(n-1)X^{n-2}Y^2d + \dots,$$

the discriminant is a function of b, c, d , &c., which can be expressed as a determinant; but the actual development is effected more conveniently by writing the quantic in the factorial form

$$bX^n + ncX^{n-1}(X+Y) + \frac{1}{2}n(n-1)dX^{n-2}(X+Y)^2 + \dots,$$

and then equating to zero the differential coefficients in regard to X and Y ; the process gives for the discriminant a product of factors which are linear functions of b, c, d , &c.

Seminvariants expressed as Determinants.

Art. Nos. 18 to 21.

18. The seminvariant 2 may be written as a determinant in the form

$$2 = \begin{vmatrix} b & c \\ 1 & b \end{vmatrix}.$$

19. The seminvariant 3 may be written as a determinant; we have

$$3 = -\frac{1}{2}(d - 3bc + 2b^3) = \frac{1}{2} \begin{vmatrix} b & c & d \\ 2 & 2b & 3c \\ 0 & 1 & b \end{vmatrix}.$$

20. And generally any seminvariant may be expressed as a determinant; the form is most easily obtained by means of the differential equation $\delta S = 0$, or $a\partial_b S + 2b\partial_c S + 3c\partial_d S + \dots = 0$, which is satisfied by the seminvariant S .

21. In particular, the seminvariant may be expressed as a determinant formed with the coefficients of the differential equation; for instance, the seminvariant 22 is the determinant

$$22 = \begin{vmatrix} b & c & d & e \\ 3 & 3b & 4c & 5d \\ 0 & 2 & 2b & 3c \\ 0 & 0 & 1 & b \end{vmatrix}.$$

The Fundamental Seminvariants. Art. Nos. 22 to 28.

22. The seminvariants 2, 3, 4, 5, ... are called the *fundamental seminvariants*; they are obtained in order by the formula

$$c = (1, b, \mathfrak{e} - b, 1)^2, \quad d = (1, b, c, \mathfrak{d} - b, 1)^3, \quad e = (1, b, c, d, \mathfrak{e} - b, 1)^4, \quad \&c.,$$

or, what is the same thing,

$$\begin{aligned} c &= b^2 - c, \\ d &= b^3 - 3bc + 2d, \\ e &= b^4 - 6b^2c + 3c^2 + 8bd - 6e, \end{aligned}$$

&c.; viz. up to g , these are

$$\begin{aligned}c &= b^2 - c, \\d &= b^3 - 3bc + 2d, \\e &= b^4 - 6b^2c + 3c^2 + 8bd - 6e, \\f &= b^5 - 10b^3c + 15bc^2 + 20b^2d - 20bd - 30cd + 24f, \\g &= b^6 - 15b^4c + 45b^2c^2 - 15c^3 + 40b^3d - 120bcd + 40d^2 \\&\quad - 90b^2e + 90ce + 144bf - 120g.\end{aligned}$$

23. These numbers are at once obtained by the formula; thus for e , multiplying successively by $-b$ and 1, we have

$$\begin{array}{c} 1, b, c, d, e \\ -b, -b^2, -bc, -bd, -be \\ 1, b, c, d \end{array}$$

and thence $e = b^4 - 6b^2c + 3c^2 + 8bd - 6e$.

24. It may be remarked that the fundamental seminvariant c is the discriminant of $(b, \zeta X, Y)^2$, d is the discriminant of $(b, c, \zeta X, Y)^3$, and so on.

25. In regard to the expression of a seminvariant as a rational and integral function of the fundamental seminvariants, it is to be observed that the expression is not unique; for instance, we have $22 = c^2 - d$, but also $22 = c^2 - bd + cd - d$, &c.; among these expressions there is one which is the most simple, and which is to be adopted as the standard form.

26. The standard form is obtained by writing the seminvariant as a function of b, c, d , &c., and then substituting for c, d, e , &c. their values in terms of the fundamental seminvariants c, d, e , &c.; the expression is then reduced by means of the relations which connect the fundamental seminvariants; for instance, from $c = b^2 - c$, we have $b^2 = c + c$, and thence $b^4 = c^2 + 2cc + c^2$, &c.

27. The standard form of a seminvariant is a rational and integral function of the fundamental seminvariants, which does not contain the highest fundamental seminvariant in a degree higher than 1; and it is further restricted by the condition that the coefficients are numerical fractions in their lowest terms.

28. It is to be observed that the number of independent seminvariants is infinite; but that for any given weight the number of seminvariants is finite, being equal to the number of non-unitary partitions of that weight.

Perpetuants. Art. Nos. 29 to 35.

29. A *perpetuant* is a seminvariant which cannot be expressed as a rational integral function of seminvariants of lower weight; for instance, the seminvariants 2, 3, 4, 5 are perpetuants, but $22 = c^2 - d$ is not a perpetuant, since it can be expressed in terms of c and d , which are of lower weights.

30. Sylvester established in like manner the notion of a *perpetuant*; a perpetuant is a seminvariant which is not expressible as a rational integral function of seminvariants of lower weight; or, what is the same thing, it is a seminvariant which is not expressible as a rational integral function of seminvariants of the same or lower degree.

31. The number of perpetuants of a given weight w is equal to the number of non-unitary partitions of w into two or more parts; or, what is the same thing, it is equal to the number of non-unitary partitions of w diminished by 1 (for the partition into a single part gives the fundamental seminvariant, which is a perpetuant). The formula for the number of perpetuants of weight w is

$$\frac{1}{w}\{2^{w-1} - 1 - (-1)^w\} - \frac{1}{2}\{1 - (-1)^w\},$$

or, what is the same thing,

$$\frac{1}{w}\{2^{w-1} - 1 - (-1)^w\} - \frac{1}{2} + \frac{1}{2}(-1)^w.$$

32. The formula may be written in the form

$$\frac{2^{w-1} - 1}{w} - \frac{1 - (-1)^w}{2},$$

and it is to be observed that the second term vanishes when w is odd, and is $= 1$ when w is even.

33. The number of perpetuants of a given weight w and degree j is given by the formula of Sylvester; the number is equal to the coefficient of x^w in the expansion of

$$\frac{x^{\frac{1}{2}j(j+1)}}{(1-x^2)(1-x^3)\dots(1-x^j)}.$$

34. In particular, the number of quadric perpetuants of weight w is the coefficient of x^w in $\frac{x^3}{1-x^2}$, which is $= 1$ when w is odd and ≥ 3 , and $= 0$ when w is even; and the number of cubic perpetuants of weight w is the coefficient of x^w in $\frac{x^6}{(1-x^2)(1-x^3)}$.

35. The generating function for the number of perpetuants is

$$\frac{x^2}{1-x^2} + \frac{x^3}{1-x^3} + \frac{x^4}{(1-x^2)(1-x^4)} + \frac{x^5}{(1-x^2)(1-x^5)} + \dots,$$

or, as given by Sylvester,

$$\frac{x^2}{1-x^2} + \frac{x^3}{(1-x^2)(1-x^3)} + \frac{x^4}{(1-x^2)(1-x^3)(1-x^4)} + \dots$$

The Seminvariants of a given Degree. Art. Nos. 36 to 40.

36. The seminvariants of a given degree are connected by linear relations; for instance, in degree 2, the seminvariants are c, d, e, f , &c., and these are connected by no linear relation; in degree 3, the seminvariants are cd, ce, de, c^3, d^2 , &c., and these are connected by linear relations.

37. The number of linearly independent seminvariants of degree j and weight w is given by the formula

$$(w; j, n) - (w - 1; j, n),$$

where $(w; j, n)$ denotes the number of partitions of w into j parts not exceeding n .

38. For the seminvariants of a quantic of infinite order, the number of linearly independent seminvariants of degree j and weight w is equal to the coefficient of x^w in the expansion of

$$\frac{x^{\frac{1}{2}j(j+1)}}{(1-x)(1-x^2)\dots(1-x^j)}.$$

39. The generating function for the number of linearly independent seminvariants of degree j is

$$\frac{x^{\frac{1}{2}j(j+1)}}{(1-x)(1-x^2)(1-x^3)\dots}.$$

40. The total number of linearly independent seminvariants of weight w (for a quantic of infinite order) is the coefficient of x^w in

$$\frac{1}{(1-x)(1-x^2)(1-x^3)\dots} - 1 - \frac{x}{1-x},$$

which is the generating function for the number of non-unitary partitions of w .

The Seminvariants of a Quantic of given Order. Art. Nos. 41 to 45.

41. For a quantic of order n , the number of seminvariants of weight w and degree j is the same as for a quantic of infinite order, provided that $w \leq n$ and $j \leq n$; but for higher values of w or j , the number is diminished.

42. The number of seminvariants of a quantic of order n is finite; the highest weight of a seminvariant is $\frac{1}{2}n(n-1)$, and the highest degree is n .

43. The generating function for the number of seminvariants of a quantic of order n is given by Sylvester's formula; it is the coefficient of z^n in the expansion of a certain rational function.

44. The seminvariants of a quantic of order n may be obtained from the seminvariants of a quantic of infinite order by equating to zero all coefficients after the $(n+1)$ th coefficient.

45. Attending next to the seminvariants of a quantic of a given order, we have to consider the relations which exist between the seminvariants of different degrees; these relations are called *syzygies*, and they form an important part of the theory.

46. A syzygy is a linear relation between products of seminvariants; for instance, for the quintic, we have the syzygy $c \cdot d - cd - c^3 = 0$, which expresses that the product $c \cdot d$ is equal to the sum of the seminvariants cd and c^3 .

47. The syzygies of a quantic of given order may be obtained by equating to zero the coefficients of the powers of x in the identity which expresses that the product of two seminvariants is a seminvariant.

48. The number of syzygies of a given weight and degree is given by a formula analogous to that for the number of seminvariants; the difference between the number of products and the number of seminvariants gives the number of syzygies.

49. In the case of a quantic of infinite order, there are no syzygies; but for a quantic of finite order, the number of syzygies increases with the order of the quantic.

50. Supposing that for any given weight $w - 6$, each of the quintic perpetuants *is* a sextic syzygant: this implies that the number of sextic syzygies $(S_6)'$ is at least equal to the number $((5))'$ of quintic perpetuants (for each expression of a quintic perpetuant as a sextic syzygant is in fact a sextic syzygy): and not only so, but it further implies that the number of the sextic syzygies, which do not contain a quintic perpetuant, is precisely equal $(S_6)' - ((5))'$: for if besides the equations which serve to express the perpetuants as syzygants, we have any other sextic syzygy, then either this does not contain a quintic perpetuant, or it can (by substituting therein for every quintic perpetuant its value as a sextic syzygant) be reduced to a syzygy which does not contain any quintic perpetuant.

51. In the general case, we have $(S_6)'$ sextic syzygies of the weight $w - 6$, and $((5))'$ quintic perpetuants of this weight: but it may happen that certain of the quintic perpetuants do not enter into any of the sextic syzygies; and those which enter, may do so in definite combinations: by elimination of these combinations of perpetuants we obtain (it may be) α sextic syzygies not containing any quintic perpetuant; and the remaining $(S_6)' - \alpha$ equations will then serve to express each of them a quintic perpetuant, or combination of quintic perpetuants, as a sextic syzygant. The number α is at most $= ((5))'$, or taking it to be $= ((5))' - ((6))'$, the number of sextic syzygies not containing any quintic perpetuant will be $= (S_6)' - ((5))' + ((6))'$, that is, the number of sextic syzygies not containing any quintic perpetuant will be equal to the whole number $(S_6)'$ of sextic syzygies diminished by some number $((5))' - ((6))'$, which is less than or at most equal to the whole number $((5))'$ of quintic perpetuants of the weight in question $w - 6$. But as already mentioned,

I have not been able to obtain the expression of the function $(6) = \frac{\omega(x)}{2.3.4.5}$, which is the G. F. of the number $((6))'$.

Cambridge, England, 17th March, 1884.

**TABLES OF THE SYMMETRIC FUNCTIONS OF
THE ROOTS,
TO THE DEGREE 10, FOR THE FORM**

$$1 + bx + \frac{cx^2}{1.2} + \dots = (1 - \alpha x)(1 - \beta x)(1 - \gamma x) \dots$$

[From the *American Journal of Mathematics*, t. VII. (1885), pp. 47—56.]

THE tables are derived from the tables (b) of my “Memoir on the Symmetric Functions of the Roots of an Equation,” *Phil. Trans.*, vol. cxlvii. (1857), pp. 489—496, [147]. These refer in effect to the form $1 + bx + cx^2 + \dots$, and we have consequently to change $b, c, d, \dots$ into $\frac{b}{1}, \frac{c}{1.2}, \frac{d}{1.2.3}, \dots$ respectively. Thus in the heading of the original table V(b), we must instead of

$$f, \quad be, \quad cd, \quad b^2d, \quad bc^2, \quad b^3c, \quad b^5,$$

write

$$\begin{aligned} & \frac{f}{120}, \quad \frac{be}{24}, \quad \frac{cd}{12}, \quad \frac{b^2d}{6}, \quad \frac{bc^2}{4}, \quad \frac{b^3c}{2}, \quad \frac{b^5}{1}, \\ &= \frac{1}{120}(f, \quad 5be, \quad 10cd, \quad 20b^2d, \quad 30bc^2, \quad 60b^3c, \quad 120b^5); \end{aligned}$$

the several columns of the original table are then multiplied by 1, 5, 10, 20, 30, 60, 120, and we thus obtain the new table with the heading

$$\frac{1}{120}(f, \quad be, \quad cd, \quad b^2d, \quad bc^2, \quad b^3c, \quad b^5).$$

In the original tables, there is a remarkable property (very easily proved) in regard to the sums of the numbers in a *column*. Thus for the table V(b) these sums are

$$-1, \quad +2, \quad +2, \quad -3, \quad -3, \quad +4, \quad -1,$$

where the sign is + or – according as the heading is the product of an even or an odd number of letters; and the numerical value depends only on the indices in the heading: these indices are

$$1, \quad 11, \quad 11, \quad 21, \quad 21, \quad 31, \quad 5,$$

and they give the foregoing values

$$1, \quad 2, \quad 2, \quad 3, \quad 3, \quad 4, \quad 1,$$

viz. $b^3c, = 31$ gives the value $\Pi 4 \div \Pi 3 \cdot \Pi 1, = 4$; b^2d, bc^2 , each $= 21$, give the value $\Pi 3 \div \Pi 2 \cdot \Pi 1, = 3$; and so in other cases.

In the new tables we have the like property; thus for the table $V(b)$, the sums are

$$-1, \quad +2, \quad +2, \quad -3, \quad -3, \quad +4, \quad -1,$$

and for the several tables the values are

Table	Sums
I(b)	+1
II(b)	-1, +1
III(b)	+1, -2, +1
IV(b)	-1, +2, +2, -3
V(b)	-1, +2, +2, -3, -3, +4, -1
VI(b)	+1, -2, -2, +3, +3, -4, +4, -4, +2, -1

and so on; the law is the same as in the original tables.

The following are the tables up to the degree 10:

I(b). $a = b$.

II(b). $a^2 = b^2 - c$.

$$\begin{array}{c|c|c} & b^2 & c \\ \hline 2 & +1 & -1 \end{array}$$

III(b). $a^3 = b^3 - 3bc + 2d$.

$$\begin{array}{c|c|c|c} & b^3 & bc & d \\ \hline 3 & +1 & -3 & +2 \end{array}$$

IV(b). $a^4 = b^4 - 6b^2c + 3c^2 + 8bd - 6e$.

$$\begin{array}{c|c|c|c|c|c} & b^4 & b^2c & c^2 & bd & e \\ \hline 4 & +1 & -6 & +3 & +8 & -6 \\ 31 & +1 & -2 & -1 & +2 & \\ 22 & +1 & -2 & +1 & & \end{array}$$

V(b). $a^5 = b^5 - 10b^3c + 15bc^2 + 20b^2d - 20bd - 30cd + 24f$.

$$\begin{array}{c|c|c|c|c|c|c|c} & b^5 & b^3c & bc^2 & b^2d & cd & be & f \\ \hline 5 & +1 & -10 & +15 & +20 & -30 & -20 & +24 \\ 41 & +1 & -4 & +3 & +4 & -6 & -2 & \\ 32 & +1 & -4 & +3 & +2 & & -2 & \\ 311 & +1 & -2 & -1 & +2 & & & \\ 221 & +1 & -2 & +1 & & & & \end{array}$$

VI(b).

	b^6	b^4c	b^2c^2	c^3	b^3d	bcd	b^2e	c^2d	bd^2	bf
6	+1	-15	+45	-15	+40	-120	-90	+40	+40	+144
51	+1	-7	+9	+3	+12	-18	-30	-6	+12	+24
42	+1	-7	+9	+3	+6	-18	-18	+6		+12
411	+1	-5	+3	+3	+4	-6	-6	-6	+4	
33	+1	-6	+6	+3	+8	-18	-18		+8	+24
321	+1	-5	+3	+3	+2	-6	-6		+2	
3111	+1	-3	-3	+3		+6		-6		
222	+1	-3	+3	-1						
2211	+1	-3	+3	-1						

VII(b).

	b^7	b^5c	b^3c^2	bc^3	b^4d	b^2cd	c^2d	b^2e	bce	b^2f	cd	bd^2	bg	h
7	+1	-21	+105	-105	+70	-420	+210	-210	+630	+504	-210	+280	-840	+720
61	+1	-11	+27	+15	+22	-66	-18	-66	+90	+72	+18	+36	-72	
52	+1	-11	+27	+15	+10	-66	-30	-30	+90	+36	+30		-36	
511	+1	-8	+9	+15	+8	-18	-18	-18	+18	+18	+18	+8	-18	
43	+1	-10	+18	+15	+16	-48	-24	-48	+72	+60	+24	+24	-60	
421	+1	-8	+9	+15	+4	-18	-18	-18	+18	+18	+18	+4	-18	
4111	+1	-5	-9	+15		+18	-18		-18		+18			
331	+1	-7	+9	+3	+8	-18	-6	-18	+18	+18	+6	+8	-18	
322	+1	-6	+9	-3	+4	-18	+6	-18	+18	+18	-6	+4	-18	
3211	+1	-5	-3	+3	+2	+6	-6	-6	-6	+6	+6	+2	-6	
31111	+1	-3	-3	+3		+6	-6		-6		+6			

VIII(*b*).

	b^8	b^6c	b^4c^2	b^2c^3	c^4	b^5d	b^3cd	bc^2d	b^4e	b^2ce	c^2e	b^2d^2	cd^2	b^3f	bce	b^2g	deg
8	+1	-28	+210	-420	+105	+112	-840	+840	-560	+1680	-420	+560	-560	+1344	-1680	-1344	+840
71	+1	-16	+60	-40	-40	+36	-168	+72	-120	+264	+72	+96	-24	+192	-168	-192	+24
62	+1	-16	+60	-40	-40	+16	-168	+120	-48	+264	-24	+24	+24	+72	-72	-72	-24
611	+1	-12	+24	-8	-40	+14	-48	+24	-40	+48	+24	+40	-24	+48	-24	-48	-24
53	+1	-15	+45	-30	-45	+22	-132	+90	-84	+198	-18	+48	+6	+108	-90	-108	-6
521	+1	-12	+24	-8	-40	+10	-48	+24	-30	+48	+24	+20	-24	+30	-24	-30	-24
5111	+1	-8	-6	+16	-8	+8	+12	-24	-8	-12	+24	+8	-8	+8	-8	-8	-8
44	+1	-14	+42	-28	-35	+24	-144	+96	-96	+216	-24	+48		+120	-96	-120	
431	+1	-11	+21	-5	-40	+12	-48	+24	-36	+48	+24	+24	-24	+36	-24	-36	-24
422	+1	-10	+24	-12	-24	+8	-60	+36	-24	+60	-12	+12	+12	+24	-24	-24	-12
4211	+1	-8	-3	+12	-12	+6	+6	-18	-12	+6	+18	+12	-12	+12	-6	-12	-12
41111	+1	-5	-9	+15	-3		+18	-18		-18	+18		+18		-18		+18
332	+1	-9	+18	-6	-15	+8	-48	+30	-24	+48	-6	+12	+6	+24	-18	-24	-6
3311	+1	-7	-3	+9	-12	+6	+6	-18	-12	+6	+18	+12	-12	+12	-6	-12	-12
3221	+1	-6	+3	+3	-9	+2	-12	+12	-6	+12	-6	+4	+6	+6	-6	-6	-6
32111	+1	-4	-6	+6	+3	+2	+6	-12	-6	-6	+12	+2	+6	+2	-6	-2	-6
311111	+1	-2	-6	+6	+3		+6	-12		-6	+12		+6		-6		-6
2222	+1	-4	+6	-4	+1						+12		+6				
22211	+1	-4	+6	-4	+1												

IX(b).

	b^9	b^7c	b^5c^2	b^3c^3	bc^4	b^6d	b^4cd	b^2c^2d	c^3d	b^5e	b^3ce	bc^2e	b^4f	b^2cf	c^2f	b^3g
9	+1	-36	+378	-1260	+945	+168	-1512	+2520	-1260	-1260	+5040	-3780	+3024	-7560	+3780	-6048
81	+1	-22	+117	-195	-117	+52	-312	+312	+78	-234	+468	+78	+312	-390	-78	-312
72	+1	-22	+117	-195	-117	+22	-312	+390	-78	-102	+468	-156	+120	-156	+78	-120
711	+1	-17	+54	-45	-117	+18	-90	+72	-18	-72	+108	+36	+72	-54	-36	-72
63	+1	-21	+99	-153	-135	+34	-198	+252	-54	-162	+378	-108	+162	-216	+54	-162
621	+1	-17	+54	-45	-117	+14	-90	+72	-18	-54	+108	+36	+54	-54	-36	-54
6111	+1	-12	-9	+45	-27	+12	+18	-54	+18	-18	-18	+54	+18	-18	-18	-18
54	+1	-20	+90	-135	-135	+28	-168	+216	-54	-132	+324	-108	+132	-180	+54	-132
531	+1	-16	+48	-39	-117	+16	-72	+60	-18	-48	+72	+36	+48	-36	-36	-48
522	+1	-15	+51	-57	-81	+10	-84	+102	-36	-36	+84	-30	+36	-42	+18	-36
5211	+1	-12	-3	+33	-39	+8	+6	-36	+18	-18	-6	+36	+18	-6	-18	-18
51111	+1	-7	-18	+30	+9	+4	+24	-36	-12	-4	-24	+36	+4	-4	-12	-4
441	+1	-14	+42	-36	-72	+16	-72	+60	-18	-48	+72	+36	+48	-36	-36	-48
432	+1	-13	+45	-51	-72	+10	-72	+84	-30	-30	+72	-24	+30	-36	+12	-30
4311	+1	-10	-3	+27	-36	+8	+6	-30	+18	-18	-6	+30	+18	-6	-18	-18
4221	+1	-9	+9	+9	-27	+4	-24	+30	-12	-12	+24	-12	+12	-12	+6	-12
42111	+1	-6	-9	+18	+9	+4	+12	-24	-6	-12	-12	+24	+12	-12	-6	-12
411111	+1	-3	-18	+18	+9	+12	-24	-24	-12	-12	-12	+24	-12	+12	+12	-24
333	+1	-12	+36	-30	-45	+8	-48	+42	-12	-24	+48	-12	+24	-24	+12	-24
3321	+1	-9	+9	+9	-27	+4	-24	+30	-12	-12	+24	-12	+12	-12	+6	-12
33111	+1	-6	-9	+18	+9	+4	+12	-24	-6	-12	-12	+24	+12	-12	-6	-12
3222	+1	-8	+18	-12	-18	+2	-24	+24	-6	-6	+18	-12	+6	-6	+6	-6
32211	+1	-5	-3	+12	-9	+2	+6	-18	+6	-6	-6	+18	+6	-6	-6	-6
321111	+1	-3	-12	+12	+9	+2	+12	-18	-6	-6	-12	+18	+6	-6	-6	-6
3111111	+1	-1	-12	+12	+9	+12	-18	-18	-6	-6	-12	+18	-6	+6	+6	-6
22221	+1	-5	+12	-10	+3											
222111	+1	-4	+6	-4	+1											

$X(b)$.

	b^{10}	b^8c	b^6c^2	b^4c^3	b^2c^4	c^5	b^7d	b^5cd	b^3c^2d	bc^3d	b^6e	b^4ce	b^2c^2e	c^3e	b^5f	b^3f^2
10	+1	-45	+630	-3150	+4725	-945	+210	-2520	+6300	-6300	-3150	+12600	-15750	+3150	+5040	-15750
91	+1	-29	+207	-429	-195	+351	+78	-546	+702	+234	-390	+780	+390	-390	+468	-390
82	+1	-29	+207	-429	-195	+351	+28	-546	+858	-78	-180	+780	-390	-78	+180	-180
811	+1	-23	+99	-117	-195	+117	+26	-156	+156	+26	-130	+156	+78	-78	+130	-130
73	+1	-28	+189	-378	-189	+378	+48	-336	+504	-126	-252	+630	-504	+126	+336	-504
721	+1	-23	+99	-117	-195	+117	+20	-156	+156	+26	-102	+156	+78	-78	+102	-102
7111	+1	-17	-9	+81	-81	+9	+16	+12	-84	+36	-36	-36	+108	-36	+36	-36
64	+1	-27	+171	-333	-189	+351	+40	-282	+432	-126	-204	+540	-432	+126	+264	-432
631	+1	-22	+90	-102	-162	+108	+18	-126	+126	+18	-90	+126	+54	-54	+90	-90
622	+1	-21	+93	-129	-135	+189	+12	-138	+192	-60	-60	+138	-120	+60	+60	-60
6211	+1	-17	-3	+69	-87	+45	+10	-6	-60	+36	-30	-6	+60	-30	+30	-30
61111	+1	-11	-21	+51	+9	-27	+6	+30	-48	-12	-6	-30	+48	+12	+6	-6
55	+1	-26	+162	-315	-189	+324	+36	-252	+396	-126	-180	+504	-414	+126	+252	-180
541	+1	-21	+84	-93	-153	+99	+16	-108	+108	+18	-72	+108	+54	-54	+72	-72
532	+1	-20	+87	-120	-126	+168	+10	-114	+162	-54	-48	+114	-102	+54	+48	-48
5311	+1	-16	-3	+63	-81	+45	+8	-6	-48	+30	-24	-6	-48	-24	+24	-24
5221	+1	-14	+12	+30	-60	+36	+4	-36	+48	-18	-12	+30	+30	+18	+12	-12
52111	+1	-9	-18	+42	+12	-24	+4	+18	-36	-6	-6	-18	+36	+6	+6	-6
511111	+1	-5	-30	+30	+30	-9	+2	+30	-36	-18	-2	-30	+36	+18	+2	-2
442	+1	-18	+72	-102	-108	+162	+8	-96	+138	-48	-36	+96	-90	+48	+36	-96
4411	+1	-14	+3	+39	-63	+36	+6	+6	-42	+24	-18	-6	+42	-24	+18	-42
433	+1	-17	+63	-81	-81	+135	+6	-72	+102	-36	-24	+72	-66	+36	+24	-66
4321	+1	-13	+12	+30	-60	+36	+4	-30	+42	-18	-12	+24	-24	+18	+12	-12
43111	+1	-8	-18	+36	+18	-27	+4	+12	-30	-6	-12	+30	+6	+6	+12	-6
4222	+1	-11	+24	-18	-36	+27	+2	-30	+36	-12	-6	+24	-24	+12	+6	-12
42211	+1	-7	-9	+27	+9	-18	+2	+6	-24	+12	-6	-6	+24	-12	+6	-6
421111	+1	-4	-24	+24	+24	-9	+2	+18	-24	-12	-6	-18	+24	+12	+6	-12
4111111	+1	-2	-30	+24	+36	-9		+18	-24	-12	-18	+24	+24	+12		
3331	+1	-12	+27	-18	-45	+27	+2	-30	+36	-12	-6	+24	-24	+12	+6	-12
3322	+1	-10	+24	-18	-36	+27	+2	-24	+30	-12	-6	+18	-18	+12	+6	-12
33211	+1	-6	-9	+24	+9	-18	+2	+6	-24	+12	-6	-6	+24	-12	+6	-6
331111	+1	-3	-24	+24	+24	-9	+2	+18	-24	-12	-6	-18	+24	+12	+6	-12
32221	+1	-7	+12	+6	-24	+18	+1	-18	+24	-9	-3	+12	-15	+9	+3	-9
322111	+1	-4	-12	+18	+18	-15	+1	+12	-18	-6	-3	-12	+18	+6	+3	-6
3211111	+1	-2	-21	+18	+27	-12	+1	+18	-18	-12	-3	-18	+18	+12	+3	-12
31111111	+1	-1	-24	+18	+30	-12		+18	-18	-12		-18	+18	+12		
22222	+1	-6	+15	-20	+15	-6										
222211	+1	-5	+9	-7	-3	+6										
2221111	+1	-3	+6	-4	+1											

$X(b)$ (continued). — The table is concluded by showing the remaining columns:

Columns: b^4d^2 , b^2cd^2 , c^2d^2 , bd^3 , b^4e , b^2ce , c^2e , bde , b^2f , cf , bg , h continued.

[The table $X(b)$ is continued with additional columns showing the symmetric functions of weight 10 with the headings b^4d^2 , b^2cd^2 , c^2d^2 , bd^3 , b^4e , b^2ce , c^2e , bde , b^2f , cf , bg , h , and the corresponding numerical coefficients for each partition.]

The law of the sums of the columns is verified for all the tables; the numerical values depend on the indices of the headings according to the same law as in the original tables.

NON-UNITARY PARTITION TABLES.

[From the *American Journal of Mathematics*, t. VII. (1885), pp. 57, 58.]

IN the theory of Seminvariants we are concerned with the non-unitary partitions of a number, that is, the number of ways of making up the number with the parts 2, 3, 4, ...; or what is the same thing, writing $\mathbf{2} = 1 - x^2$, $\mathbf{3} = 1 - x^3$, &c., with the Generating Functions having in their denominators the factors $\mathbf{2}$, $\mathbf{3}$, $\mathbf{4}$, &c. In the present short paper, I give the developments up to x^{100} of the functions $1 \div \mathbf{2}$, $\mathbf{2.3}$, $\mathbf{2.3.4}$, $\mathbf{2.3.4.5}$, $\mathbf{2.3.4.5.6}$, respectively: and also of the function

$$x^6 + x^{13} - 2x^{16} - x^{18} + x^{31} \div \mathbf{2.3.4.5.6},$$

which function is (there is strong reason to believe) the G. F. for the number of sextic syzygies of a given weight: the same function without the term x^{31} occurs (p. 115) in Professor Sylvester's paper "On Subinvariants, i.e. Seminvariants to Binary Quantics of an Unlimited Order," *American Journal of Mathematics*, t. V. (1882), pp. 79—136.

In the tables, X is written to denote $x^6 + x^{13} - 2x^{16} - x^{18} + x^{31}$.

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NON-UNITARY PARTITION TABLES.

Ind. x	$\overset{1\div}{2.3}$	2.3.4	2.3.4.5	2.3.4.5.6	$\overset{X\div}{2.3.4.5.6}$	Ind. x	$\overset{1\div}{2.3}$	2.3.4	2.3.4.5	2.3.4.5.6	$\overset{X\div}{2.3.4.5.6}$
0	1	1	1	1		50	9	65	258	750	186
1	0	0	0	0		51	9	61	268	783	226
2	1	1	1	1		52	9	70	286	854	203
3	1	1	1	1		53	9	65	297	891	248
4	1	2	2	2		54	10	75	316	972	223
5	1	1	2	2		55	9	70	328	1010	270
6	2	3	3	4	1	56	10	80	348	1098	242
7	1	2	3	3	0	57	10	75	361	1144	294
8	2	4	5	6	1	58	10	85	382	1236	262
9	2	3	5	6	1	59	10	80	396	1287	319
10	2	5	7	9	2	60	11	91	419	1391	284
11	2	4	7	9	2	61	10	85	433	1443	344
12	3	7	10	14	4	62	11	96	457	1555	306
13	2	5	10	13	4	63	11	91	473	1617	371
14	3	8	13	19	6	64	11	102	598	1734	328
15	3	7	14	20	7	65	11	96	515	1802	399
16	3	10	17	26	8	66	12	108	541	1932	353
17	3	8	18	27	11	67	11	102	559	2002	427
18	4	12	22	36	13	68	12	114	587	2142	377
19	3	10	23	36	15	69	12	108	606	2223	457
20	4	14	28	47	17	70	12	120	635	2369	402
21	4	12	29	49	21	71	12	114	655	2457	490
22	4	16	34	60	22	72	13	127	686	2618	429
23	4	14	36	63	28	73	12	120	707	2709	519
24	5	19	42	78	29	74	13	133	739	2881	456
25	4	16	44	80	35	75	13	127	762	2985	552
26	5	21	50	97	36	76	13	140	795	3164	483
27	5	19	53	102	44	77	13	133	819	3276	586
28	5	24	60	120	43	78	14	147	854	3472	513
29	5	21	63	126	54	79	13	140	879	3588	620
30	6	27	71	149	53	80	14	154	916	3797	542
31	5	24	74	154	64	81	14	147	942	3927	656
32	6	30	83	180	62	82	14	161	980	4144	572
33	6	27	87	189	78	83	14	154	1008	4284	693
34	6	33	96	216	72	84	15	169	1048	4520	604
35	6	30	101	227	89	85	14	161	1077	4665	730
36	7	37	111	260	84	86	15	176	1118	4915	636
37	6	33	116	270	102	87	15	169	1149	5076	769
38	7	40	127	307	96	88	15	184	1192	5336	568
39	7	37	133	322	117	89	15	176	1224	5508	809
40	7	44	145	361	108	90	16	192	1269	5789	703
41	7	40	151	378	133	91	15	184	1302	5967	849
42	8	48	164	424	123	92	16	200	1349	6264	736
43	7	44	171	441	149	93	16	192	1384	6460	891
44	8	52	185	492	137	94	16	208	1432	6768	772
45	8	48	193	515	167	95	16	200	1469	6977	934
46	8	56	207	568	152	96	17	217	1519	7308	809
47	8	52	216	594	186	97	16	208	1557	7524	977
48	9	61	232	656	169	98	17	225	1609	7873	846
49	8	56	241	682	205	99	17	217	1649	8109	1022
						100	17	234	1883	8651	883

SEMINVARIANT TABLES.

[From the *American Journal of Mathematics*, t. VII. (1885), pp. 59—73.]

THE present tables are not, I think, superseded by the tables A, pp. 149—163, contained in Capt. MacMahon's paper, "Seminvariants and Symmetric Functions," *American Journal of Mathematics*, t. VI. (1883), pp. 131—163. His order of the terms, though a very ingenious one, and giving rise to a most remarkable symmetry in the form of the tables, seems to me too artificial—and I cannot satisfy myself that it ought to be adopted in preference to the more simple one which I use: I attach also considerable importance to the employment of the simple letters b, c, d, e , &c. in place of the suffixed ones a_1, a_2, a_3, a_4 , &c. There is, moreover, the question of the identification of the seminvariants with their expressions as non-unitary symmetric functions of the roots of the equation $1 + bx + \frac{cx^2}{1.2} + \&c. = 0$, which requires to be considered.

As to the form in which the tables present themselves, I remark that every seminvariant is a rational and integral function of the fundamental seminvariants

$$\begin{aligned} c &= (1, b, \& - b, 1)^2, \\ d &= (1, b, c, \& - b, 1)^3, \\ e &= (1, b, c, d, \& - b, 1)^4, \quad \&c., \end{aligned}$$

viz. up to g , these are

$c =$	$d =$	$e =$	$f =$	$g =$	$ce =$	$d^2 =$	$c^3 =$
$c + 1$	$d + 1$	$e + 1$	$f + 1$	$g + 1$			
$b^2 - 1$	$bc - 3$	$bd - 4$	$be - 5$	$bf - 6$	+1	+1	
	$b^3 + 2$	c^2	cd	ce	-1	-6	+1
		$b^2c + 6$	$b^2d + 10$	$b^2e + 15$	-4	+9	-3
		$b^4 - 3$	bc^2	bcd	+4	+4	+3
			$b^3c - 10$	$b^3d - 20$	+6	-12	-1
			$b^5 + 4$	b^2c^2	-9	+4	
				$b^4c + 15$	+3		
				$b^6 - 5$			

and if to the value of any seminvariant expressed in terms of b, c, d , &c. we substitute the foregoing values of c, d, e , &c., the result will be the value of the seminvariant in terms of the fundamental seminvariants c, d, e , &c.

It will be observed that, as regards the terms b^2c, bc^2 , &c., the coefficients in the several columns are the binomial coefficients with the proper signs; this is a general law.

SEMINVARIANT TABLES UP TO ($m = 12$).

($b = 1$)

($c = 2$)

($d = 3$)

b	c	d
+1	+1	+1

($e = 4$)

e	c^2	bd	b^2c
+1	+1	-4	+2
b	-1	+3	-1

($f = 5$)

($g = 6$)

f	ce	d^2	c^2d	b^2e	bcd	b^3d	bc^2	b^4c
+1	+1	+1	-1	+2	-5	-5	+5	-2
b	-1	-6	+2	-1	+4	+3	-2	+1
c		+9	-3		-3		+1	
b^2		-3	+1		+1		-1	

$(h = 7)$

h	cg	df	e^2	c^2f	cde	d^3	c^3d	b^2g	bce	b^2f	bd^2	...
+1	+1	+1	+1	-1	-1	+1	+1	+2	-6	-6	+9	...
b	-1	-7	-8	+2	+3	+3	-2	-1	+5	+4	-6	...
c		+9	+20	-3	-6	-12	+3		-4	-3	+9	...
d		-5	-12	+1	+3	+9	-1		+1	+1	-4	...
b^2		-3	-8	+1	+2	+6	-1		+1	+1	-3	...
bc			+4		-1	-4			-1	-1	+2	...
b^3			+1			+1					-1	...

 $(i = 8)$

i	ch	dg	ef	c^2g	cdf	ce^2	d^2f	c^3e	cd^2	b^2h	bcg	bd^2	be^2	...
+1	+1	+1	+1	-1	-1	-1	+1	+1	+1	+2	-7	-7	-8	...
b	-1	-8	-10	+2	+3	+4	+4	-2	-3	-1	+6	+5	+6	...
c		+10	+25	-3	-7	-16	-20	+3	+9		-5	-4	-12	...
d		-6	-16	+1	+4	+10	+16	-1	-6		+1	+1	+4	...
e		+1	+4		-1	-4	-8		+1				-1	...
b^2		-4	-12	+1	+3	+6	+8	-1	-3		+1	+1	+3	...
bc			+6		-1	-4	-8		+3		-1	-1	-2	...
bd			-1			+1	+2		-1					...
b^3			+1				-1							...

$(j = 9)$

j	ci	dh	eg	f^2	c^2h	cdg	cef	df^2	d^2e	c^3g	c^2df	cde^2	cd^2e	d^4	$\dots$
+1	+1	+1	+1	+1	-1	-1	-1	-1	+1	+1	+1	+1	-1	+1	...
b	-1	-9	-12	-14	+2	+3	+4	+5	+5	-2	-3	-4	+2	-3	...
c		+11	+30	+42	-3	-8	-19	-30	-36	+3	+9	+18	-6	+12	...
d		-7	-22	-35	+1	+5	+14	+26	+36	-1	-6	-15	+6	-15	...
e		+2	+8	+16		-1	-5	-12	-20		+1	+6	-1	+6	...
f			-1	-4			+1	+3	+6			-1		-1	...
b^2		-5	-16	-24	+1	+4	+10	+15	+18	-1	-4	-10	+4	-9	...
bc			+8	+16		-2	-8	-16	-24		+3	+12	-6	+15	...
bd			-2	-6		+1	+3	+8	+14		-1	-5	+2	-7	...
be				+1			-1	-2	-4			+1		+1	...
b^3			+1	+2			-1	-2	-3			+1	-1	+2	...

The tables are continued for higher values; the law of formation is as follows: the coefficients in each column are the binomial coefficients with the proper signs, and the terms are arranged in the order of the powers of b , c , d , &c.

($k = 10$) continued:

k	cj	di	eh	fg	c^2i	cdh	ceg	cf^2	d^2g	def	e^3	...
+1	+1	+1	+1	+1	-1	-1	-1	-1	+1	+1	+1	...
b	-1	-10	-14	-18	+2	+3	+4	+5	+6	+6	+7	...
c		+12	+35	+54	-3	-9	-22	-38	-30	-48	-72	...
d		-8	-28	-52	+1	+6	+18	+38	+24	+48	+96	...
e		+2	+11	+28		-1	-7	-19	-8	-22	-56	...
f			-2	-8			+1	+5	+1	+6	+22	...
g				+1				-1		-1	-6	...

($l = 11$) continued:

l	ck	dj	ei	fh	g^2	c^2k	cdj	cei	cfh	cg^2	d^2i	dfh	...
+1	+1	+1	+1	+1	+1	-1	-1	-1	-1	-1	+1	+1	...
b	-1	-11	-16	-22	-26	+2	+3	+4	+5	+6	+7	+7	...
c		+13	+40	+68	+90	-3	-10	-25	-46	-66	-42	-60	...
d		-9	-34	-70	-105	+1	+7	+22	+50	+90	+36	+66	...
e		+2	+14	+38	+70		-1	-9	-27	-60	-12	-30	...
f			-3	-12	-30			+1	+7	+22	+2	+10	...
g				+2	+8				-1	-5		-1	...
h					-1					+1			...

($m = 12$) continued:

m	cl	dk	ej	fi	gh	c^2l	cdk	cej	cfi	cgh	d^2j	dfi	...
+1	+1	+1	+1	+1	+1	-1	-1	-1	-1	-1	+1	+1	...
b	-1	-12	-18	-26	-32	+2	+3	+4	+5	+6	+7	+8	...
c		+14	+45	+84	+120	-3	-11	-28	-54	-84	-56	-84	...
d		-10	-40	-92	-150	+1	+8	+27	+63	+120	+48	+96	...
e		+2	+17	+51	+100		-1	-11	-36	-84	-16	-48	...
f			-4	-18	-48			+1	+9	+30	+2	+14	...
g				+3	+12				-1	-7		-2	...
h					-1					+1			...

[The tables are continued in the same manner for higher values, the law of formation being always the same: the coefficients are the binomial coefficients with the proper signs, and the terms are arranged in the order of the powers of b, c, d, e, f, g , &c.]